



# HASAN NASRALLAH

SENIOR AI ENGINEER · AGENTIC & GEOSPATIAL AI

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## 01 · SUMMARY

AI Engineer with 7+ years of experience designing and deploying end to end machine learning systems, from ML ready datasets and model training to scalable cloud delivery. Now focused on **agentic AI**: building autonomous agents that pair LLMs such as Claude with **MCP tool servers, knowledge bases and document RAG** to run real world workflows at production cost. Specialized in Earth observation, remote sensing and satellite imagery.

## 02 · EXPERIENCE

### Lead AI Engineer

RASID.AI · Lebanon

2023 → PRESENT

- Architected **Rasid GoPilot**, an agentic AI platform for satellite imagery analysis that plans and runs complete Earth observation workflows, built with Claude on AWS Bedrock AgentCore
- Built **GoServer**: 100+ geospatial tools across domain MCP servers (data access, vector, raster and AI inference) on serverless GPUs, with plan based tool access
- Cut agent cost per request by **up to 99%** (about 89% on typical tasks) through dynamic tool search, on demand tool loading, prompt caching and a 67% leaner system prompt
- Extending the suite with **GoBase**, a geospatial knowledge base (PostGIS, DuckDB, Dagster), **GoDoc**, a document RAG pipeline for reports and maps, and **GoClient**, a Python SDK
- Designed a scalable Geo AI engine serving 13 segmentation, detection and embedding models, including SAM3 and DINOv3
- Built and deployed models for urban building and change detection, crop field delineation, wheat monitoring, and crop health and stress analysis
- Built large scale datasets and trained models for global methane emission monitoring using real and synthetic satellite imagery, enabling detection and alerting workflows

### AI Engineer

Orbis Holding · Lebanon

2020 → 2023

- Developed and optimized computer vision models for face detection, tracking, recognition, emotion analysis, and liveness detection using CNNs, GANs, and advanced optimization in Python and C++
- Built AI powered OCR and document parsing systems for automated ID and passport verification in real world environments such as airports and retail
- Designed models for face image quality assessment, blur detection, and forgery detection in identity documents to strengthen verification accuracy and security

### Research Assistant

National Council for Remote Sensing, Lebanon (CNRS)

2018 → 2020

- Developed end to end AI pipelines for building detection from satellite imagery, covering ML ready dataset creation, model design, training, and large scale deployment
- Generated a nationwide urban map of Lebanon and designed a solar energy estimation algorithm to evaluate photovoltaic potential across buildings
- Built a large scale dataset and trained deep learning models for silo bin detection from Mapbox satellite imagery

## 03 · EDUCATION

### Masters in Computer and Communication Engineering

Lebanese University

2015 → 2020 · Beirut, Lebanon

7+

YEARS EXP

4

PAPERS

20+

PROJECTS

## RASID GOPILOT

SEEING EARTH, SMARTER

Flagship agentic AI platform I lead at RASID: an agent with 100+ MCP tools, 13 GPU AI models and 10+ data sources.

GoServer  
MCP SERVERS

GoBase  
KNOWLEDGE BASE

GoDoc  
DOCUMENT RAG

GoClient  
PYTHON SDK

▶ [app.rasid.ai/try-gopilot](https://app.rasid.ai/try-gopilot)

## SKILLS

### AGENTIC AI

AI Agents · MCP · Tool Search · RAG · Knowledge Bases · Prompt Caching · Context Engineering · AWS Bedrock AgentCore

### MACHINE LEARNING

Computer Vision · Deep Learning · PyTorch · Vision Language Models · Foundation Models · LLMs · MLOps

### PROGRAMMING, CLOUD & DATA

Python · C/C++ · Java · AWS · GCP · Serverless GPUs · PostGIS · DuckDB · Dagster

### REMOTE SENSING

QGIS · ArcGIS · Google Earth Engine · GDAL

## KEY ACHIEVEMENTS

### ◆ Up to 99% Lower Agent Cost

Re engineered GoPilot's tool use with dynamic tool search

### ◆ ML Competitions

Developed solutions for NASA Harvest, SpaceNet, Open Cities, Xview2, and BirdCLEF 2022

## PUBLICATIONS

[Zero Shot Refinement of Buildings Segmentation Models using SAM](#)  
ARXIV

[Empirical Study of PEFT Techniques for Winter Wheat Segmentation](#)  
ARXIV

[Scale Invariant Model for Building Detection from Aerial Images](#)  
SPRINGER

[Lebanon Solar Rooftop Potential Assessment via Buildings Segmentation](#)  
IEEE